

Higher-Order Metaphysics
 Week Five: Engaging with some skeptics
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 18th October 2023

First... some unfinished business from last week!

1 Extended types and terms

The *extended type system* $\mathcal{T}^+$ is the smallest set that contains e , t , $\perp$, and $\top$, and such that it contains $(\sigma \rightarrow \tau)$, $(\sigma \wedge \tau)$, and $(\sigma \vee \tau)$ whenever it contains σ and τ .¹

A *extended signature* Σ is a collection of strings (constants) plus an assignment to them of extended types; an *extended environment* is a finite list of variable, extended type pairs. Given such a signature, the *extended language* $\mathcal{L}(\Sigma)$ is the smallest set of $\langle$ extended environment, string, extended type $\rangle$ triples meeting the following conditions:²

$$\begin{array}{c}
 \frac{\sigma \in \mathcal{T}^+}{\Gamma, v : \sigma \vdash v : \sigma} \text{Var} \quad \frac{\Gamma \vdash A : \sigma \quad \tau \in \mathcal{T}^+}{\Gamma, v : \tau \vdash A : \sigma} \text{AddVar} \quad \frac{c :_{\Sigma} \sigma}{\vdash c :_{\Sigma} \sigma} \text{Const} \\
 \frac{\Gamma, v : \sigma \vdash A : \tau}{\Gamma \vdash (\lambda v^{\sigma}. A) : \sigma \rightarrow \tau} \rightarrow I \quad \frac{\Gamma \vdash A : \sigma \rightarrow \tau \quad \Gamma \vdash B : \sigma}{\Gamma \vdash (AB) : \tau} \rightarrow E \\
 \frac{\Gamma \vdash A : \sigma \quad \Gamma \vdash B : \tau}{\Gamma \vdash \langle A, B \rangle : \sigma \wedge \tau} \wedge I \quad \frac{\Gamma \vdash A : \sigma \wedge \tau}{\Gamma \vdash \pi_1 A : \sigma} \wedge E1 \quad \frac{\Gamma \vdash A : \sigma \wedge \tau}{\Gamma \vdash \pi_2 A : \tau} \wedge E2 \\
 \frac{\Gamma \vdash A : \sigma \quad \tau \in \mathcal{T}}{\Gamma \vdash l_{\tau}^1 A : \sigma \vee \tau} \vee I1 \quad \frac{\Gamma \vdash B : \tau \quad \sigma \in \mathcal{T}}{\Gamma \vdash l_{\sigma}^2 B : \sigma \vee \tau} \vee I2 \\
 \frac{\Gamma \vdash A : \sigma \vee \tau \quad \Gamma, u : \sigma \vdash B : \rho \quad \Gamma, v : \tau \vdash C : \rho}{\Gamma \vdash \delta A(u.B)(v.C) : \rho} \vee E \\
 \frac{}{\Gamma \vdash * : \top} \top I \quad \frac{\Gamma \vdash A : \perp \quad \sigma \in \mathcal{T}}{\Gamma \vdash \varepsilon_{\sigma} A : \sigma} \perp E
 \end{array}$$

¹ A common alternative notation uses 0 , 1 , $+$ and $\times$ instead of $\perp$, $\top$, $\vee$ and $\wedge$.

² We could perfectly well set this up in the old way where variables have built-in types and there is no need to relativize to an environment: the relativization just helps display the analogy with intuitionistic propositional logic.

We can also consider languages that add some but not all of the four new type-forming operations.

2 β - and η -equivalence and reduction for extended types

To do any interesting *logical* work with our new terms we will need some rules for manipulating them, analogous to β - and η -conversion.

These rules will correspond to rules of *proof simplification*: the β -rules correspond to getting rid of an introduction rule followed immediately by the corresponding elimination rule, and the η -rules are the other way around.

Just like most syntactic notions, the relevant equivalence relations $\sim_\beta$, $\sim_{\eta'}$, $\sim_{\beta\eta}$ need to be relativized to a context Γ —we write $\Gamma \vdash A \sim B$.³

$$\begin{array}{c}
\frac{\Gamma \vdash A : \sigma \quad \Gamma \vdash B : \tau}{\Gamma \vdash \pi^1 \langle A, B \rangle \sim A} \beta_\wedge^1 \qquad \frac{\Gamma \vdash A : \sigma \quad \Gamma \vdash B : \tau}{\Gamma \vdash \pi^2 \langle A, B \rangle \sim B} \beta_\wedge^2 \\
\\
\frac{\Gamma \vdash A : \sigma \wedge \tau}{\Gamma \vdash \langle \pi^1 A, \pi^2 A \rangle \sim A} \eta_\wedge \qquad \frac{\Gamma \vdash A : \top}{\Gamma \vdash * \sim A} \eta_* \\
\\
\frac{\Gamma \vdash A : \sigma \quad \Gamma \vdash B : \tau \quad \Gamma \vdash C : \tau}{\Gamma \vdash \delta(\iota_\tau^1 A)(u . B)(v . C) \equiv B[A/u]} \beta_\vee^1 \qquad \frac{\Gamma \vdash A : \sigma \quad \Gamma \vdash B : \tau \quad \Gamma \vdash C : \tau}{\Gamma \vdash \delta(\iota_\sigma^2 A)(u . B)(v . C) \equiv C[A/v]} \beta_\vee^2 \\
\\
\frac{\Gamma \vdash A : \sigma \vee \tau \quad \Gamma \vdash B : \rho}{\Gamma \vdash \delta A(u . B[\iota_\tau^1 u/x])(v . B[\iota_\sigma^2 v/x]) \sim B[A/x]} \eta_\vee \qquad \frac{\Gamma \vdash A : \perp \quad \Gamma \vdash B : \sigma}{\Gamma \vdash \varepsilon_\sigma A \sim B} \eta_\perp
\end{array}$$

³ In the case of the simple type system, such relativization turns out not to be needed—terms that are equivalent relative to one environment are equivalent relative to any environment in which both are typeable terms. But in the extended system (in the setup where variables lack built in types) the relativization is crucial: if it were dropped, the rules for $*$ and ε would reduce $\sim$ to triviality.

3 Logics for extended types

The signature of higher order logic can be extended naturally to the new types by allowing quantifiers $\forall_\alpha$, $\exists_\alpha$ and identity constants $=_\alpha$ in the new types. Given this, the definitions of the logics H, C, and E carry over straightforwardly to the extended system, plugging in the extended notion of $\beta\eta$ -conversion instead of the old one.

But there are two wrinkles, a major one involving the empty type $\perp$ and a minor wrinkle involving sum-types.

4 Logics that allow for there being nothing

Given how we defined H, all formulae of the form $\exists x^\sigma . x =_\sigma x$ are theorems. This would be a disaster if we allowed the empty type $\perp$ with its conversion rule!

There's a simple fix that becomes very natural in this setting. Instead of talking about the relation $\Gamma \vdash P$ (' Γ proves P '), we relativize to an environment Δ . Read $\Delta; \Gamma \vdash P$ as ' Γ proves P relative to Δ '; its truth will require Δ to declare a type for all free variables of Γ and P .

For most of the logical inference rules we just carry along Δ from the input sequent to the output. But, for $\forall$ Intro (and $\exists$ Elim), we drop the

relevant variable:

$$\frac{\Delta, v : \sigma; \Gamma \vdash P \quad v \notin FV(\Gamma)}{\Delta; \Gamma \vdash \forall v^\sigma. P} \text{ } \forall\text{Intro}$$

The resulting “existence-neutral” version of H doesn’t prove $\exists x^e. x =_e x$, though it does still prove $\exists p^t. p =_t p$, and indeed $\exists x^\tau. x =_\tau x$ where τ is any type ending in t . This can be carried over reasonably into a type system with $\perp$ as a type (and its conversion rule $\eta_\perp$).

5 A minor wrinkle involving sum-types

There’s one less important wrinkle: in the logic H so defined, we can’t prove that everything in a sum-type is derived by injection from something in one of the summands:

$$\forall x^{\alpha \vee \beta}. (\exists y^\alpha. x = \iota_\beta^1 \alpha) \vee (\exists z^\beta. x = \iota_\alpha^2 \beta)$$

We *can* prove this in (extended) Classicism; but it seems so fundamental that we’d probably want it even if we weren’t fans of Classicism in general. So it’s natural to add it as an axiom; or equivalently (and more elegantly in a natural deduction setting), the following new rule:

$$\frac{\Gamma \vdash P[\iota_\beta^1 u^\alpha / z^{\alpha \vee \beta}] \quad \Gamma \vdash P[\iota_\alpha^2 v^\beta / z^{\alpha \vee \beta}] \quad z \text{ free for } A \text{ in } P \quad u^\alpha, v^\beta \notin FV(\Gamma \cup \{P\})}{\Gamma \vdash P[A / z^{\alpha \vee \beta}]} \text{ Sum-Subst}$$

6 Normal forms for extended types

In the setting of implicational logic [simple type theory], a crucial feature of proofs [terms] in normal form is the *subformula* [*subtype*] property: the conclusion [type] of every subproof [subterm] is either a subformula [subtype] of the conclusion [type] of the proof [term], a subformula [subtype] of one of the assumptions [types of the variables], or a subformula [subtype] of an axiom [type of a constant].

Prawitz and Gentzen showed (independently) that we can define a notion of normal form for the extended language/type system that has the same property.

This means that if we don’t have any new *constants* in the extended types, there is a sense in which we don’t get anything new by introducing the extended types: every *sentence* (in fact, every simply-typed closed term) in the extended language will be $\sim_{\beta\eta}$ -equivalent to [hence synonymous with?] something in the old language.

If we *do* have constants floating around in the new types, we can still define a sort of “translation” from the extended language to an old lan-

guage by first systematically replacing each such constant with an extended expression built up from only simply-typed constants, and then translating.

For example, if we have an extended constant a of type $e \wedge e$, we could first replace it with $\langle b, c \rangle$ where b and c are new type- e constants, and then reduce to normal form.

7 Eliminating the extended quantifiers

What about the *logical* constants? First consider the simple case of a language with only products and $\top$. Here, there is an extremely natural strategy defining the extended quantifiers, along the following lines:⁴

⁴ This isn't a complete definition—you also need to say what $\forall_{\gamma \rightarrow (\alpha \vee \beta)}$ is, etc. But it's not hard to fill in the rest of the details.

$$\begin{aligned}\forall_{\alpha \wedge \beta} &:= \lambda X^{(\alpha \wedge \beta) \rightarrow t}. \forall y^\alpha z^\beta. X \langle y, z \rangle \\ \forall_\top &:= \lambda X^{\top \rightarrow t}. X * \\ \forall_{(\alpha \wedge \beta) \rightarrow \gamma} &:= \lambda X^{((\alpha \wedge \beta) \rightarrow \gamma) \rightarrow t}. \forall Y^{\alpha \rightarrow \beta \rightarrow \gamma}. X(\lambda z^{\alpha \wedge \beta}. Y \pi^1 z \pi^2 z) \\ \forall_{\top \rightarrow \gamma} &:= \lambda X^{(\top \rightarrow \gamma) \rightarrow t}. \forall Y^\gamma. X(\lambda z^\top. Y)\end{aligned}$$

We can prove in extended H that these are logically equivalent to the primitive quantifiers of these types.⁵

⁵ And so we can prove in extended Classicism that they are identical.

What if we let in sum-types?

Can this carry over to sum types? Well...sometimes.

Let N and T be the smallest pair of extended types such that:

$$\frac{}{e \in N} \quad \frac{}{t \in T} \quad \frac{}{\perp \in N} \quad \frac{}{\top \in T} \quad \frac{\alpha \in (N \cup T) \quad \beta \in T}{(\alpha \rightarrow \beta) \in T} \quad \frac{\alpha \in N \quad \beta \in N}{(\alpha \wedge \beta) \in N} \quad \frac{\alpha \in T \quad \beta \in T}{(\alpha \wedge \beta) \in T} \quad \frac{\alpha \in N \cup T \quad \beta \in N \cup T}{(\alpha \vee \beta) \in N}$$

Then, there are natural definitions of $\forall_\alpha$ for all $\alpha \in N \cup T$ which are provably logically equivalent to the originals (in H extended with Sum-Subst). For example:

$$\begin{aligned}\forall_{\alpha \vee \beta} &:= \lambda X^{(\alpha \vee \beta) \rightarrow t}. (\forall y^\alpha. X l_\beta^1 \alpha) \wedge (\forall z^\beta. X l_\alpha^2 \beta) \\ \forall_{(\alpha \vee \beta) \rightarrow \gamma} &:= \lambda X^{((\alpha \vee \beta) \rightarrow \gamma) \rightarrow t}. \forall y^{\alpha \rightarrow \gamma} z^{\beta \rightarrow \gamma}. X(\lambda x^{\alpha \vee \beta}. \delta x(u . yu)(v . zv))\end{aligned}$$

8 Hofweber

The thesis of Hofweber 2022 is that the ‘results of HOL’ lack significance for “metaphysics”, since they do not answer the question ‘what reality is like’—and in fact do not answer any question that can be asked in English!

Initially, he seems willing to concede that one could use HOL to answer *other* questions, not statable in English: he says that ‘as far as I am concerned there is nothing wrong with higher-order languages’ (34). But eventually it emerges that the concession is supposed to apply equally to ‘Rune English’ which supplements English with Viking runes used as sentential operators, introduced simply by talking about syntax and model theory. He is in fact skeptical that the questions of HOL have answers that are ‘simply true full stop, not merely true in this or that model’; that they lack content ‘full stop’;; that they are ‘meaningful in the ordinary sense’, and ‘have content’.

He distinguishes ‘analyzers’, who ‘find the resources of HOL in natural languages’, from ‘augmenters’, who ‘hold that the relationship between HOL and natural language is in the end insignificant, since we can simply add the resources of HOL to natural language if they aren’t already there, and then benefit from the added expressive strength in our meta-physical theorizing’.

9 Hofweber versus ‘analyzers’

In English, expressions formed from sentences by prefixing ‘that’ or ‘the fact that’ or ‘the proposition that’ are, obviously, not syntactically interchangeable with sentences. Sentences can go syntactically in places where they can’t go, and they can go syntactically in places where sentence can’t go.⁶

- (1) (a) That snow is white is well-known
- (b) * Snow is white is well-known
- (c) * That that snow is white is well-known

Hofweber allows that despite this, one might hold that ‘on the proper semantic analysis’ quantifiers ‘into’ the position of such expressions will turn out to be quantifiers into sentence position. However, ‘there is a simple and straightforward observation that quite conclusively shows that quantification into that-clause position, and with it quantification over facts, is not sentential quantification, nor any other kind of higher-order quantification in English.’

The argument concerns sentences like

⁶ From the syntactic point of view, ‘the fact that P’ seems interchangeable with any other definite description. ‘That’-clauses are different:

- (a) It is surprising [that France has no king/*the king of France/*the fact that France has no king].
- (b) I thought about [the king of France/the fact that France has no king/*that France has no king]

(18): Something is disgusting.

According to Hofweber, this has a single reading where it is implied by all of the following sentences:

- (2) (a) The mess John left behind is disgusting.
- (b) The fact that John is the new head of mess enforcement is disgusting.
- (c) It is disgusting that John is the new head of mess enforcement.

H anticipates that someone might respond that (18) is ambiguous: it has one reading where it is implied by (2a) and a different reading where it is implied by (2b) and (2c), but no reading where it is implied by both. In response, he considers:

19: There are two things that bother John: his ulcer and the fact that I won at chess.

Here there is only one occurrence of a quantifier-word, so a response in terms of ambiguity doesn't seem to get off the ground.

10 *Aside: other candidates for quantification into non-standard syntactic positions in English*

The strongest candidate for a quantifier-word that's associated with an entirely different English syntactic category are the words 'wherever', 'whenever', and 'however', also 'somehow', 'somewhere', and 'some-time' in sentences like

- (3) John danced however Mary danced.
- (4) John went wherever Mary went.
- (5) John sneezed whenever Mary sneezed.⁷

⁷ 'However Mary danced', etc., are *adverbial phrases*.

You can do quite a lot with 'however': e.g., Rayo and Yablo (2001) argue that sentences like 'Ann and Bob are related however Carol and Dan' = are related' are English equivalents of $\forall R^{e \rightarrow e \rightarrow t}. Rab \rightarrow Rcd$.

More complicated are uses of 'something' and 'everything' as in the following sentences:

- (6) He is something that I am not—kind.⁸

⁸ Prior 1971: 37.

The thought is that ‘something that I am not’ is itself of the same grammatical category as ‘kind’ (an adjectival phrase), since this meaning of ‘is’ requires a complement of this category.⁹ However, as Moltmann 2013 points out, these quantifiers can combine with adjectives in a way that is tricky to understand on this view:

(7) He is something important—kind.

Given that ‘kind is important’ is ungrammatical,

11 *Hofweber versus the ‘augmenters’ part 1: can HOL answer any English?*

Can any sentence of HOL (that is not also a sentence of FOL) be an *answer* to any English question? Hofweber thinks not, and concludes that sentences of HOL are not relevant to English questions like ‘Is reality structured?’ or even ‘What is the case?’

He has a grammatical reason for this: in English, a full, non-elliptical answer to the question ‘What is the case?’ is a sentence of some special form such as ‘It is the case that snow is white’. Ordinary sentences like ‘snow is white’ are only elliptical answers. Even though he is willing to grant that it is sometimes possible to answer a question in one language in a different language, he sees no reason to think that any sentence of HOL is an answer to ‘what is the case’, or indeed any other English question!

- One thing I’m confused about is why Hofweber isn’t open to a view on which certain questions askable in English are answerable in an *extended* form of English that includes the symbolism of HOL as well as other English expressions. In such a language, one could write ‘It is the case that...’ in front of some higher-order sentence such as RM (the RM theorem), and that would be an answer to the question ‘what is the case?’.¹⁰

Suppose we grant H that ‘snow is white’ isn’t a non-elliptical answer to ‘what is the case?’. Still it’s presumably an answer to ‘what is snow like?’.¹¹ Similarly, any sentence with a proper name *N* in subject position seems to be a non-elliptical answer to the question ‘What is *N* like?’.

This suggests that if a HO language contains a term *a* synonymous with some English proper name, any sentence of the form *Fa* should be an answer to the question ‘what is *a* like?’. It shouldn’t matter whether *F* is expressible in English—even $(\lambda x.RM)a$ should count as an answer.

⁹ Of course, ‘something that I am not’ cannot go everywhere ‘kind’ can go:

* He is a [kind/*something that I am not] person

¹⁰ Of course, one might reject the inference from RM itself to the result of prefixing ‘it is the case that’ to it, maybe on the grounds of some kind of sparse theory of facts.

¹¹ Someone might deny this I suppose, saying that any non-elliptical answer to this would have to be of the form ‘Snow is like...’. The problem is that ‘Snow is white’ must surely be an *elliptical* answer to ‘What is snow like?’, but it’s not obvious what sentence of that form it would be elliptical for!

12 Hofweber versus ‘augmenters’ part 2: Answering questions in HOL

Suppose the conclusion of the previous section is right: no HOL sentence is an answer to any English question. One might nevertheless be curious about questions asked in a higher-order language (supplemented with question marks). What’s the problem?

According to H, ‘for the revisionary approach to HOM to make sense, it should lead to a better question for us to ask. It should lead to a question which is more suitable for the project of metaphysics than the English Target’. He complains that there is no reason for thinking that the HOL questions are any better than the English ones, and that given their infelicity in English, it could not be rational for us to ‘abandon’ a perfectly meaningful English question in favour of one we can’t represent in English.

- Something odd is going on here. Why would investigating questions posed in a higher-order language require abandoning any English questions? Couldn’t we continue to investigate them simultaneously?¹²

¹² I suppose one might worry that spending time on the HOL questions would slow progress on the English ones; but (a) sometimes new questions turn out to help with old ones, and (b) for better or worse, many of us are lucky to have jobs where we can investigate whatever questions we find interesting, without having to convince anyone else that they are interesting.

References

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